

DFMEA - t6_r1_singlemodel (G2_kappa0175)

Document: VBF-T6R1SINGLEMODEL-DFMEA-001

Failure mode	Effect	S	O	D	RPN	Mitigation / status
Lithium plating at the 4C charge (the anode potential < 0)	the capacity loss, the dendrite risk	9	3	2	54	DESIGNED OUT: the early-trip razor trips the 4.7 V limit at $t_{ch} \sim 65$ s with the anode surface potential min +0.0454 V (the 45 mV margin). The residual risk: the margin is thin by the production-tolerance standards - the BMS must enforce the 4.7 V cutoff strictly.
Thermal over-limit (the $T > 323.15$ K)	the degradation acceleration	7	2	2	28	MITIGATED: the h 400.0 W/m ² .K cooling holds the 4C T_{max} at 321.442 K (the 1.71 K margin).
SEI growth > 500 nm	the capacity fade	6	2	2	24	MITIGATED: the suppressed SEI kinetics (1e-13 m/s) -> 133.56 nm after the 100 cycles (the 366 nm margin).
Voltage plateau < 4.1 V	the device incompatibility	6	2	2	24	MITIGATED: the midpoint 4.2078 V (the 0.108 V margin).
The charge-end state ambiguity (the cathode kinetic saturation discharge)	the under-utilization of the anode	3	5	2	30	ACCEPTED (the honest consequence): the discharge ends at the $0.73 \times Q_c$ cathode-saturation point.
The 4C fill ~1.3%	the user-facing fast-charge expectation	4	5	3	60	ACCEPTED + DISCLOSED: the voltage-limited 4C charge delivers ~0.07 Ah; the datasheet states it explicitly.